

## LECTURE 13 SPURIOUS REGRESSION, TESTING FOR UNIT ROOT

### Spurious regression

In this section, we consider the situation in which one unit root process, say  $Y_t$ , is regressed on another unit root process, say  $X_t$ , while the two processes are unrelated. Assume that

- $X_t = X_{t-1} + u_t, X_0 = 0.$
- $Y_t = Y_{t-1} + v_t, Y_0 = 0.$
- $\begin{pmatrix} u_t \\ v_t \end{pmatrix} = C(L) \varepsilon_t.$
- $\{\varepsilon_t\}$  is i.i.d.,  $E[\varepsilon_t] = 0, E[\varepsilon_t \varepsilon_t^\top] = \Sigma$  a positive definite matrix:

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}.$$

- $\sum_{j=1}^{\infty} j^{1/2} \|C_j\| < \infty, C(1)$  is nonsingular.

We can also set the initial values of  $X$  and  $Y$  to be different from zero:  $X_0 = o_p(n^{1/2})$ , and  $Y_0 = o_p(n^{1/2})$ . Let

$$\begin{aligned} \Omega &= C(1) \Sigma C(1)^\top \\ &= \begin{pmatrix} \omega_u^2 & \omega_{uv} \\ \omega_{uv} & \omega_v^2 \end{pmatrix}, \end{aligned}$$

where  $\omega_u^2$  and  $\omega_v^2$  are the long-run variances of  $u_t$  and  $v_t$  respectively, and  $\omega_{uv}$  is the long-run covariance between  $u_t$  and  $v_t$ .

Let  $\hat{\beta}_n$  be the OLS regression coefficient:

$$\hat{\beta}_n = \frac{\sum_{t=1}^n X_t Y_t}{\sum_{t=1}^n X_t^2}.$$

Here we consider the regression without an intercept; however, essentially the same results can be obtained for the regression with an intercept.

Let

$$\begin{aligned} W_{X,n}(r) &= n^{-1/2} X_{[nr]}, \\ W_{Y,n}(r) &= n^{-1/2} Y_{[nr]}. \end{aligned}$$

From the FCLT we have

$$\begin{aligned} \begin{pmatrix} W_{X,n}(r) \\ W_{Y,n}(r) \end{pmatrix} &\Rightarrow \begin{pmatrix} B_X(r) \\ B_Y(r) \end{pmatrix} \\ &= \Omega^{1/2} \begin{pmatrix} W_X(r) \\ W_Y(r) \end{pmatrix}, \end{aligned}$$

where  $W_X(r)$  and  $W_Y(r)$  are two independent standard Brownian motions. Convergence is joint, which is important for all subsequent results.

Consider first

$$\begin{aligned} n^{-3/2} \sum_{t=1}^n X_{t-1} &= n^{-1} \sum_{t=1}^n W_{X,n} \left( \frac{t-1}{n} \right) \\ &= n^{-1} \left( 0 + W_{X,n} \left( \frac{1}{n} \right) + W_{X,n} \left( \frac{2}{n} \right) + \dots + W_{X,n} \left( \frac{n-1}{n} \right) \right). \end{aligned}$$

The function  $W_{X,n}(r)$  is cadlag, constant on the interval  $(t-1)/n \leq r < t/n$ , and, therefore,

$$\begin{aligned} n^{-1} W_{X,n} \left( \frac{t-1}{n} \right) &= \int_{(t-1)/n}^{t/n} dr W_{X,n} \left( \frac{t-1}{n} \right) \\ &= \int_{(t-1)/n}^{t/n} W_{X,n}(r) dr, \end{aligned}$$

so that

$$\begin{aligned} 0 &= \int_0^{1/n} W_{X,n}(r) dr, \\ n^{-1} W_{X,n} \left( \frac{1}{n} \right) &= \int_{1/n}^{2/n} W_{X,n}(r) dr, \\ &\dots \\ n^{-1} W_{X,n} \left( \frac{n-1}{n} \right) &= \int_{(n-1)/n}^1 W_{X,n}(r) dr. \end{aligned}$$

Hence,

$$\begin{aligned} n^{-3/2} \sum_{t=1}^n X_{t-1} &= \sum_{t=1}^n \int_{(t-1)/n}^{t/n} W_{X,n}(r) dr \\ &= \int_0^1 W_{X,n}(r) dr \\ &\rightarrow_d \int_0^1 B_X(r) dr. \end{aligned}$$

The last result follows from the CMT. Next,

$$\begin{aligned} n^{-3/2} \sum_{t=1}^n X_t &= n^{-3/2} \sum_{t=1}^n (X_{t-1} + u_t) \\ &= n^{-3/2} \sum_{t=1}^n X_{t-1} + n^{-3/2} \sum_{t=1}^n u_t \\ &= n^{-3/2} \sum_{t=1}^n X_{t-1} + o_p(1) \\ &\rightarrow_d \int_0^1 B_X(r) dr. \end{aligned}$$

Now, consider

$$\begin{aligned}
n^{-2} \sum_{t=1}^n X_t^2 &= n^{-2} \sum_{t=1}^n (X_{t-1} + u_t)^2 \\
&= n^{-2} \sum_{t=1}^n X_{t-1}^2 + 2n^{-2} \sum_{t=1}^n X_{t-1} u_t + n^{-2} \sum_{t=1}^n u_t^2. \\
&= n^{-2} \sum_{t=1}^n X_{t-1}^2 + 2n^{-2} \sum_{t=1}^n X_{t-1} u_t + o_p(1).
\end{aligned}$$

First,

$$\begin{aligned}
n^{-2} \sum_{t=1}^n X_{t-1}^2 &= n^{-1} \sum_{t=1}^n \left( n^{-1/2} X_{t-1} \right)^2 \\
&= \int_0^1 W_{X,n}^2(r) dr \\
&\rightarrow_d \int_0^1 B_X^2(r) dr.
\end{aligned}$$

Next,

$$2X_{t-1}u_t = X_t^2 - X_{t-1}^2 - u_t^2,$$

and

$$\begin{aligned}
n^{-1} \sum_{t=1}^n X_{t-1} u_t &= n^{-1} \sum_{t=1}^n (X_t^2 - X_{t-1}^2 - u_t^2) / 2 \\
&= \left( n^{-1} X_n^2 - n^{-1} \sum_{t=1}^n u_t^2 \right) / 2 \\
&= \left( W_{X,n}^2(1) - n^{-1} \sum_{t=1}^n u_t^2 \right) / 2 \\
&\rightarrow_d (B_X^2(1) - \sigma_u^2) / 2 \\
&= (\omega_u^2 W_X^2(1) - \sigma_u^2) / 2,
\end{aligned}$$

where  $\sigma_u^2 = E[u_t^2]$ . Therefore,

$$n^{-2} \sum_{t=1}^n X_{t-1} u_t = o_p(1),$$

and

$$n^{-2} \sum_{t=1}^n X_t^2 \rightarrow_d \int_0^1 B_X^2(r) dr. \quad (1)$$

Next, consider  $\sum_{t=1}^n X_t Y_t$ . First,

$$\begin{aligned}
n^{-2} \sum_{t=1}^n X_{t-1} Y_{t-1} &= n^{-1} \sum_{t=1}^n \left( n^{-1/2} X_{t-1} \right) \left( n^{-1/2} Y_{t-1} \right) \\
&= \int_0^1 W_{X,n}(r) W_{Y,n}(r) dr \\
&\rightarrow_d \int_0^1 B_X(r) B_Y(r) dr.
\end{aligned}$$

Next,

$$\begin{aligned} n^{-2} \sum_{t=1}^n X_t Y_t &= n^{-2} \sum_{t=1}^n (X_{t-1} + u_t) (Y_{t-1} + v_t) \\ &= n^{-2} \sum_{t=1}^n X_{t-1} Y_{t-1} + n^{-2} \sum_{t=1}^n X_{t-1} v_t + n^{-2} \sum_{t=1}^n Y_{t-1} u_t + n^{-2} \sum_{t=1}^n u_t v_t. \end{aligned}$$

Now,  $n^{-1} \sum_{t=1}^n u_t v_t \rightarrow_p \sigma_{uv}$ , where  $\sigma_{uv} = E[u_t v_t]$ , and, therefore,

$$n^{-2} \sum_{t=1}^n u_t v_t = o_p(1).$$

Further,

$$\begin{aligned} n^{-2} \left| \sum_{t=1}^n Y_{t-1} u_t \right| &\leq n^{-2} \sqrt{\sum_{t=1}^n Y_{t-1}^2 \sum_{t=1}^n u_t^2} \\ &= n^{-1/2} \sqrt{n^{-2} \sum_{t=1}^n Y_{t-1}^2 n^{-1} \sum_{t=1}^n u_t^2} \\ &= o_p(1). \end{aligned}$$

Similarly,

$$n^{-2} \sum_{t=1}^n X_{t-1} v_t = o_p(1).$$

Thus,

$$\begin{aligned} n^{-2} \sum_{t=1}^n X_t Y_t &= n^{-2} \sum_{t=1}^n X_{t-1} Y_{t-1} + o_p(1) \\ &\rightarrow_d \int_0^1 B_X(r) B_Y(r) dr. \end{aligned} \tag{2}$$

The convergence in distribution results in (1) and (2) are joint, and it follows that

$$\begin{aligned} \hat{\beta}_n &\rightarrow_d \frac{\int_0^1 B_X(r) B_Y(r) dr}{\int_0^1 B_X^2(r) dr} \\ &= \xi. \end{aligned}$$

The result holds even if  $\{u_t\}$  and  $\{v_t\}$  are independent. One might expect that  $\hat{\beta}_n$  would converge in probability to zero; however, it converges in distribution to a random variable  $\xi$  and, therefore, is inconsistent. The random variable  $\xi$  can be interpreted as a regression coefficient from the "population" or "continuous-time" regression of the Brownian motion  $B_Y$  against  $B_X$ .

Next, consider the usual  $t$ -statistic for  $H_0 : \beta = 0$ :

$$t_{\hat{\beta}_n} = \hat{\beta}_n / \left( \frac{s_n^2}{\sum_{t=1}^n X_t^2} \right)^{1/2},$$

where  $s_n^2$  is the sample variance of the fitted residuals.

$$\begin{aligned}
s_n^2 &= (n-1)^{-1} \sum_{t=1}^n \left( Y_t - \hat{\beta}_n X_t \right)^2 \\
&= (n-1)^{-1} \left( \sum_{t=1}^n Y_t^2 + \hat{\beta}_n^2 \sum_{t=1}^n X_t^2 - 2\hat{\beta}_n \sum_{t=1}^n X_t Y_t \right) \\
&= \frac{n}{n-1} n^{-1} \left( \sum_{t=1}^n Y_t^2 - \hat{\beta}_n \sum_{t=1}^n X_t Y_t \right),
\end{aligned}$$

and, therefore,

$$\begin{aligned}
n^{-1} s_n^2 &\rightarrow_d \int_0^1 B_Y^2(r) dr - \xi \int_0^1 B_X(r) B_Y(r) dr \\
&= \int_0^1 (B_Y(r) - \xi B_X(r))^2 dr.
\end{aligned}$$

Lastly,

$$\begin{aligned}
n^{-1/2} t_{\hat{\beta}_n} &= \hat{\beta}_n / \left( \frac{n^{-1} s_n^2}{n^{-2} \sum_{t=1}^n X_t^2} \right)^{1/2} \\
&\rightarrow_d \xi / \left( \frac{\int_0^1 (B_Y(r) - \xi B_X(r))^2 dr}{\int_0^1 B_X^2(r) dr} \right)^{1/2}.
\end{aligned}$$

We conclude that

$$t_{\hat{\beta}_n} = O_p(n^{1/2}).$$

Hence, as  $n \rightarrow \infty$ , for any  $K > 0$

$$\Pr\left(|t_{\hat{\beta}_n}| > K\right) \rightarrow 1,$$

and the econometrician will reject  $H_0 : \beta = 0$  with probability approaching 1. This is despite the fact that the two variables  $X$  and  $Y$  can be independent.

## Testing for unit root

Suppose that the scalar process  $\{X_t\}$  satisfies the following assumptions:

- $X_t = \rho X_{t-1} + u_t$ ,  $X_0 = 0$ .
- $u_t = C(L) \varepsilon_t$ .
- $\{\varepsilon_t\}$  is i.i.d.,  $E[\varepsilon_t] = 0$ ,  $E[\varepsilon_t^2] = \sigma^2$ .
- $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$ ,  $C(1) \neq 0$ .

We are interested in testing

$$H_0 : \rho = 1.$$

against

$$H_1 : |\rho| < 1.$$

Under the null,  $X_t = I(1)$ , while under the alternative, it is a stationary short memory process.

Consider the regression of  $X_t$  against  $X_{t-1}$ :

$$\begin{aligned}\hat{\rho}_n &= \frac{\sum_{t=1}^n X_{t-1} X_t}{\sum_{t=1}^n X_{t-1}^2} \\ &= \rho + \frac{\sum_{t=1}^n X_{t-1} u_t}{\sum_{t=1}^n X_{t-1}^2}.\end{aligned}$$

From the previous section, we know that

$$\begin{aligned}n^{-2} \sum_{t=1}^n X_{t-1}^2 &\rightarrow_d \int_0^1 B_X^2(r) dr \\ &= \omega_u^2 \int_0^1 W_X^2(r) dr,\end{aligned}$$

where  $W_X$  is a standard Brownian motion, and

$$\begin{aligned}n^{-1} \sum_{t=1}^n X_{t-1} u_t &\rightarrow_d \frac{1}{2} (B_X^2(1) - \sigma_u^2) \\ &= \frac{1}{2} (\omega_u^2 W_X^2(1) - \sigma_u^2) \\ &= \frac{\omega_u^2}{2} (W_X^2(1) - 1) + \lambda_u,\end{aligned}$$

where

$$\begin{aligned}\lambda_u &= \frac{\omega_u^2 - \sigma_u^2}{2} \\ &= \frac{1}{2} \sum_{h=-\infty}^{\infty} (\gamma_u(h) - \sigma_u^2) \\ &= \sum_{h=1}^{\infty} \gamma_u(h).\end{aligned}$$

Now, under  $H_0 : \rho = 1$ ,

$$\begin{aligned}n(\hat{\rho}_n - 1) &= \frac{n^{-1} \sum_{t=1}^n X_{t-1} u_t}{n^{-2} \sum_{t=1}^n X_{t-1}^2} \\ &\rightarrow_d \frac{\omega_u^2 (W_X^2(1) - 1) / 2 + \lambda_u}{\omega_u^2 \int_0^1 W_X^2(r) dr} \\ &= \frac{(W_X^2(1) - 1) / 2 + \lambda_u / \omega_u^2}{\int_0^1 W_X^2(r) dr}.\end{aligned}$$

In the unit root case, the asymptotic distribution depends on functionals of a standard Brownian motion and the nuisance parameters,  $\lambda_u$  and  $\omega_u^2$ . The convergence rate of  $\hat{\rho}_n$  is faster than the usual  $\sqrt{n}$ . Next, consider the  $t$  statistic for  $H_0 : \rho = 1$ .

$$T = (\hat{\rho}_n - 1) / \left( \frac{\hat{\sigma}_u^2}{\sum_{t=1}^n X_{t-1}^2} \right)^{1/2},$$

where

$$\begin{aligned}
\hat{\sigma}_u^2 &= n^{-1} \sum_{t=1}^n \hat{u}_t^2 \\
&= n^{-1} \sum_{t=1}^n (X_t - \hat{\rho}_n X_{t-1})^2 \\
&= n^{-1} \sum_{t=1}^n (X_t - X_{t-1} - (\hat{\rho}_n - 1) X_{t-1})^2 \\
&= n^{-1} \sum_{t=1}^n u_t^2 + n (\hat{\rho}_n - 1)^2 n^{-2} \sum_{t=1}^n X_{t-1}^2 - 2 (\hat{\rho}_n - 1) n^{-1} \sum_{t=1}^n X_{t-1} u_t \\
&= n^{-1} \sum_{t=1}^n u_t^2 + O_p(n^{-1}) \\
&\rightarrow_p \sigma_u^2.
\end{aligned}$$

Thus,

$$\begin{aligned}
T &= n (\hat{\rho}_n - 1) / \left( \frac{\hat{\sigma}_u^2}{n^{-2} \sum_{t=1}^n X_{t-1}^2} \right)^{1/2} \\
&\rightarrow_d \left( \frac{(W_X^2(1) - 1) / 2 + \lambda_u / \omega_u^2}{\int_0^1 W_X^2(r) dr} \right) / \left( \frac{\sigma_u^2}{\omega_u^2 \int_0^1 W_X^2(r) dr} \right)^{1/2} \\
&= \frac{\omega_u (W_X^2(1) - 1) / 2 + \lambda_u / \omega_u^2}{\sigma_u \left( \int_0^1 W_X^2(r) dr \right)^{1/2}}.
\end{aligned}$$

Again, the asymptotic distribution of the statistic depends on the unknown nuisance parameters  $\lambda_u$ ,  $\sigma_u$  and  $\omega_u$ . Phillips (1987) and Phillips and Perron (1988) suggested an adjustment, which leads to an asymptotic distribution free of nuisance parameters. Let  $\hat{\sigma}_u^2$  and  $\hat{\lambda}$  be consistent estimators of  $\sigma_u^2$  and  $\lambda$ , where  $\lambda$  can be estimated using the Newey-West type estimator:

$$\hat{\lambda} = \sum_{h=1}^{m_n} \left( 1 - \frac{h}{m_n + 1} \right) n^{-1} \sum_{t=h+1}^n \hat{u}_t \hat{u}_{t-h}.$$

A consistent estimator of the long-run variance  $\omega_u^2$  is

$$\hat{\omega}_u^2 = \hat{\sigma}_u^2 + 2\hat{\lambda}.$$

Consider the following modification of the  $t$  statistic.

$$Z_T = \frac{\hat{\sigma}_u}{\hat{\omega}_u} T - \frac{\hat{\lambda}}{\hat{\omega}_u (n^{-2} \sum_{t=1}^n X_{t-1}^2)^{1/2}}.$$

Under  $H_0 : \rho = 1$ ,

$$Z_T \rightarrow_d \frac{1}{2} \frac{W_X^2(1) - 1}{\left( \int_0^1 W_X^2(r) dr \right)^{1/2}}.$$

Under the alternative,  $|\rho| < 1$ , and  $\hat{\rho}_n - 1$  converges in probability to a negative constant. Consequently, under the stationary alternatives,  $T$  and  $Z_T$  diverge to  $-\infty$ . One should reject the null hypothesis of a unit root when

$$Z_T < c_\alpha,$$

where  $c_\alpha$  is such that

$$\Pr(Z_T < c_\alpha) \xrightarrow{H_0: \rho=1} \alpha$$

Under the null, the distribution is non-standard. However, it is parameter-free, and the critical values can be simulated as follows. First, one generates  $n$  independent  $N(0, 1)$  random variables  $u_{1,r}^*, \dots, u_{n,r}^*$  and computes

$$Z_{T,r}^* = \frac{1}{2} \frac{(n^{-1/2} \sum_{t=1}^n u_{t,r}^*)^2 - 1}{\left(n^{-2} \sum_{t=1}^n \left(\sum_{s=1}^t u_{s,r}^*\right)^2\right)^{1/2}}.$$

One repeats this for  $r = 1, \dots, R$ , where  $R$  is large. The simulated critical value  $c_{\alpha,R}$  is the  $\alpha$  quantile of  $\{Z_{T,1}^*, \dots, Z_{T,R}^*\}$ .

While the distribution of  $Z_T$  is free of nuisance parameters, it depends on the model. For example, one would typically want to allow for an intercept,  $X_t = \mu + X_{t-1} + u_t$ . In this case,  $\hat{\rho}_n$  depends on the demeaned  $X_t$ :

$$\begin{aligned} \hat{\rho}_n &= \frac{\sum_{t=1}^n (X_{t-1} - \bar{X}_n) X_t}{\sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2}, \text{ where} \\ \bar{X}_n &= n^{-1} \sum_{t=1}^n X_{t-1}. \end{aligned}$$

We have

$$\begin{aligned} n^{-2} \sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2 &= \int_0^1 \left( W_{X,n}(r) - \int_0^1 W_{X,n}(r) dr \right)^2 dr \\ &\rightarrow_d \int_0^1 \left( B_X(r) - \int_0^1 B_X(r) dr \right)^2 dr \\ &= \omega_u^2 \int_0^1 \left( W_X(r) - \int_0^1 W_X(s) ds \right)^2 dr \\ &= \omega_u^2 \int_0^1 \widetilde{W}_X^2(r) dr, \end{aligned}$$

where  $\widetilde{W}_X$  is a demeaned standard Brownian motion:

$$\widetilde{W}_X(r) = W_X(r) - \int_0^1 W_X(s) ds.$$

Hence, in this case,

$$\begin{aligned} T &= (\hat{\rho}_n - 1) / \left( \frac{\hat{\sigma}_u^2}{\sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2} \right)^{1/2}, \\ Z_T &= \frac{\hat{\sigma}_u T}{\hat{\omega}_u} - \frac{\hat{\lambda}}{\hat{\omega}_u \left( \sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2 \right)^{1/2}}, \end{aligned}$$

and the asymptotic distribution of  $Z_T$  under the null hypothesis of a unit root is given by

$$\frac{1}{2} \frac{\widetilde{W}_X^2(1) - 1}{\left( \int_0^1 \widetilde{W}_X^2(r) dr \right)^{1/2}}.$$